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$$\begin{aligned}\int_1^{+\infty} \frac{1}{x^4} dx &= \lim_{a \rightarrow +\infty} \int_1^a \frac{1}{x^4} dx = \lim_{a \rightarrow +\infty} \left[-\frac{1}{3x^3} \right]_1^a \\ &= \lim_{a \rightarrow +\infty} \left(-\frac{1}{3a^3} \right) + \frac{1}{3} = \frac{1}{3}\end{aligned}$$

References